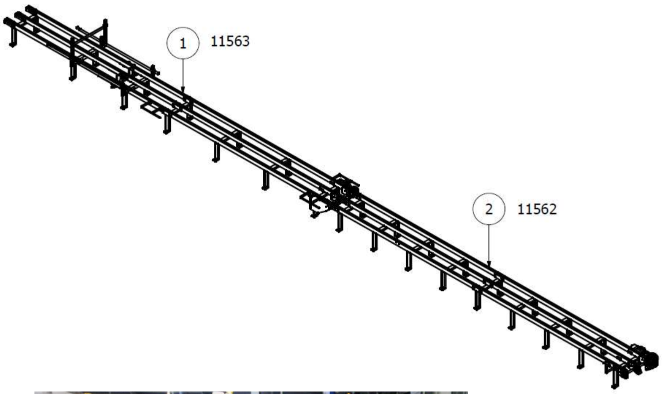
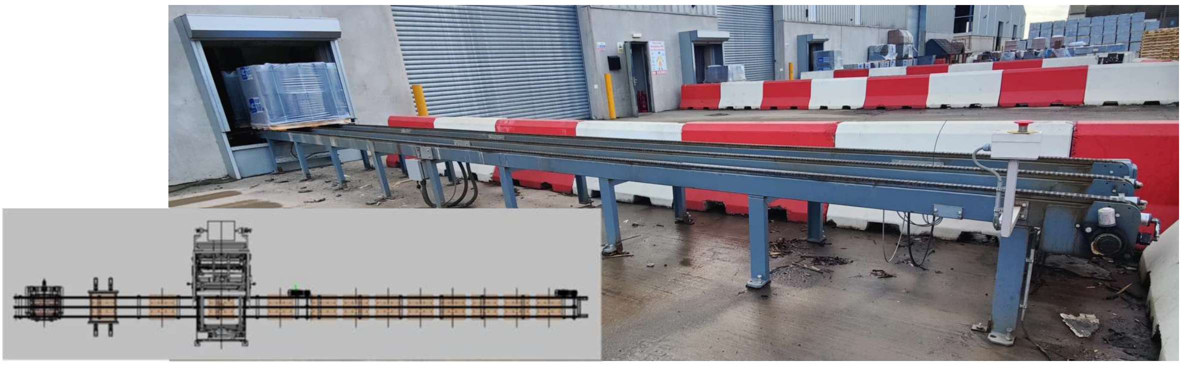
VORTEX HYDRA	**STANDARD OPERATING PROCEDURE**	Code: SOP M.02246.00 Date: 15/012/25 Page: 1 de 5
AUTOMATIC PALLETIZING LINE-Relationship SOPs		
MACHINE TYPE: PK-106 SERIAL NUMBER :M.02246		
FP McCANN (Cadeby)	ORDER : C22004	YEAR OF PRODUCTION 2023





SOP M.02246.00 Contents List Automatic Palletizing Line
SOP M.02246.01 Transport Chain 1-L.12000-M.11563
SOP M.02246.02 Transport Chain 2-L.11000-M.11562
SOP M.02246.03 Woden Pallets Automatic Dispenser
SOP M.02246.04 Squeezer

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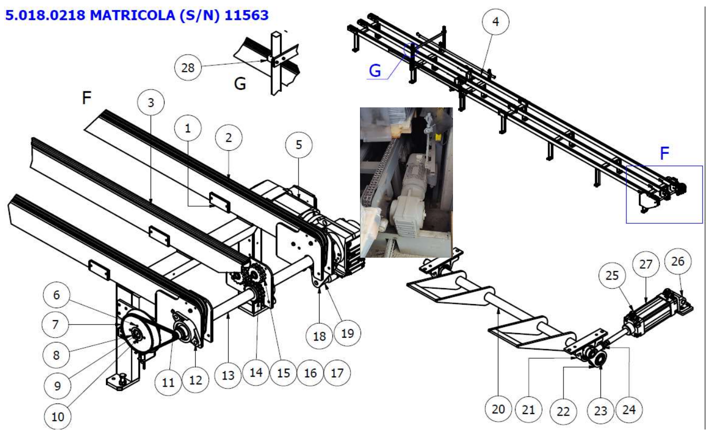
AUTOMATIC PALLETIZING LINE

MACHINE TYPE: PK-106 SERIAL NUMBER :M.02246

Transport Chain 1-L.12000-M.11563



General Description and Function

***This conveyor is the main system for forming tile pallets.**
***The process begins at the automatic wooden pallet dispenser. The empty pallet is positioned in the forming area, equipped with bottom stoppers on the conveyor that align the wooden pallet for the formation of the 3 or 6 bundles. Here, the palletizing robot deposits the tile bundles to build the load. Subsequently, the load move to the *Squeezer* (compactor) zone, which pushes the side bundles (2 or 4) toward the center to provide stability.**
***Finally, the formed and compacted pallet advances to the shrink-wrapping station, where a plastic bag is placed for shrinking and final compaction, leaving the tile ready for distribution.**

Critical Points and Risk Management

Chain Tension and Integrity:
***The point of greatest mechanical stress occurs when the robot places the heavy tile bundles onto the pallet in the forming area. Incorrect tension accelerates wear and can cause failures.**
Alignment and Condition of the Forming System:
***The conveyor's bottom stoppers must be perfectly aligned and secure.**
***A misaligned stopper causes incorrect pallet placement , leading to an unstable load and risk of damage to the conveyor itself during compaction.**
Operation of the *Squeezer* (Compactor):
Correct pressure, stroke, and alignment of the compactor cylinders are essential. Malfunction may fail to consolidate the load properly or damage the tile bundles.
Condition of Sensors and Detectors:
***The absence or failure of photocells and detectors in the forming area interrupts the automated sequence, stopping production and potentially causing collisions or jams.**

SUPERVISION LEVEL

WEELKY

1.2201.036 Chain (25 m)
2.1403.018 Connecting link for chain
C.24679 Subchain
C.24553 Subchain
2.1621.003 Bearing
2.175.013 Support
C.25059 Drive Shaft
2.174.002 Support
2.1503.035 Pneumatic Cylinder
288749 Miniaturized photocell (3 units)
252094 Inductive proximity switch (1 u)
252355 Axial reflection photoelectric switch (5 units)
277217 Scaning photocell (6 units)
18378 Limit switch (2 units)

Verification of chain tension and wear, condition of and supports, and cleaning/verification of all optical and proximity sensors.

MONTHLY

C.13739 Lever
2.2041.005 Chain Tensioner (6 units)
C.20635 Sprockets (3 units)
2.1765.106 Gearmotor + Electric Motor (1 u)

Adjust and verify the tension of all conveyor chains.

Lubrication:

***Apply grease to the bearings of all wheels according to manufacturer specifications.**
***Check levels in automatic systems.**

Area Inspection:
***Look for oil contamination on the maintenance platform in the gearbox area**

CLEANING

WEEKLY

***Clean the exterior surfaces of the gearmotor and the chains, using brushes or dry cloths, as appropriate for general ambient dust in the area.**

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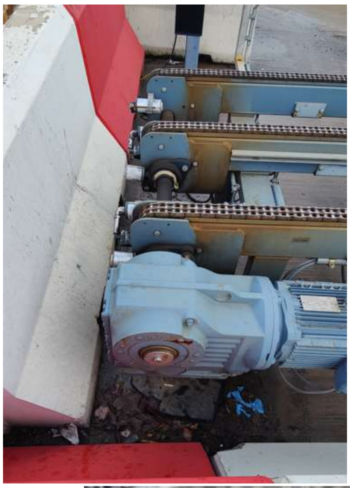
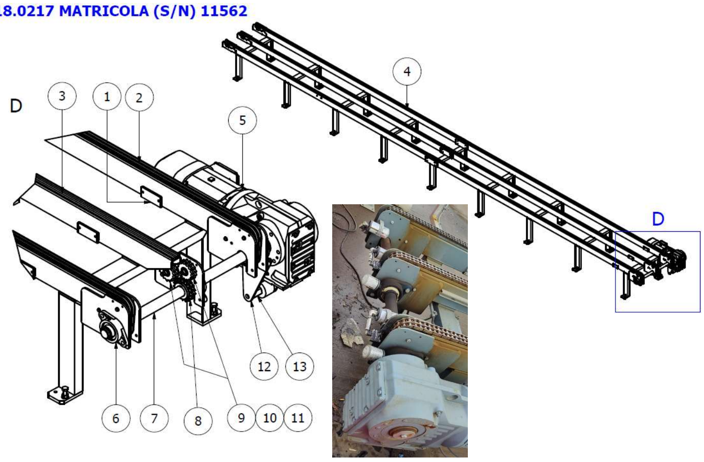
AUTOMATIC PALLETIZING LINE

MACHINE TYPE: PK-106 SERIAL NUMBER :M.02246

Transport Chain 2-L.11000-M.11562



5.018.0217 MATRICOLA (S/N) 11562



General Description and Function

*This chain conveyor receives the already formed and shrink-wrapped tile pallets from the upstream palletizing line. Its main function is to move and safely position them at the end of the line, where a forklift operator picks them up for transfer to the outdoor storage yard.
*Being located outdoors, it is directly exposed to environmental conditions.

Critical Points and Risk Management

Environmental Exposure:
*Outdoor placement subjects the chains, guides, sprockets ,and bearings to accelerated wear and deterioration due to dust, moisture, thermal changes and potential contamination.

Interaction with Forklift:
*During pallet pickup, there is a risk of accidental pressure and impacts from the forklift tynes against the conveyor structure, chains, or side components, which can cause mechanical damage or misalignments.

Chain Tension and Integrity:
Moving loaded palets under variable conditions requires correct and constant chain tension. Inadequate tension accelerates wear on links and sprockets.

SUPERVISION LEVEL

DAILY

*The forklift operator should visually observe chain sections during the day for obvious problems with connections, damaged links, or misalignments.

WEELKY

1.2201.036 Chain (23 m)
2.1403.018 Connecting link for chain
C.24580 Subchain
C.24581 Subchain
2.1621.003 Bearing
2.175.013 Support
18378 Limit switch (1 u)

Visual inspection for wear, deformed links, and lubrication.

MONTHLY

2.2041.005 Chain Tensioner (6 units)
C.20635 Sprockets (3 units)
2.1765.107 Gearmotor + Electric Motor (1 u)

Adjust and verify the tension of all conveyor chains.

Lubrication:
*Apply grease to the bearings of all wheels according to manufacturer specifications.
*Check levels in automatic systems.

Area Inspection:
*Look for oil contamination on the maintenance platform in the gearbox area

CLEANING

WEEKLY

*Clean the exterior surfaces of the gearmotor and the chains, using brushes or dry cloths, as appropriate for general ambient dust in the area.

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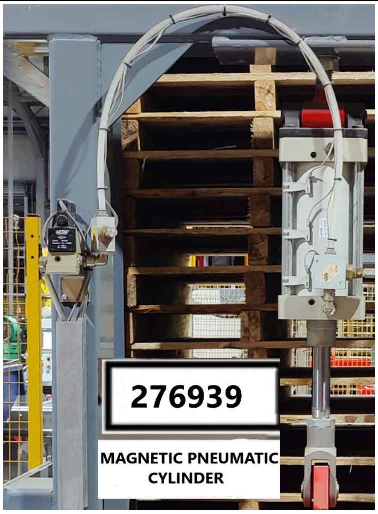
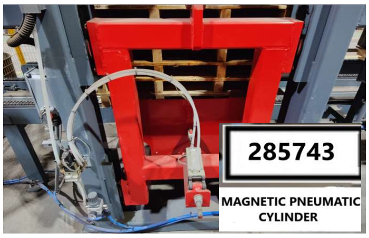
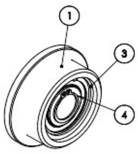
AUTOMATIC PALLETIZING LINE

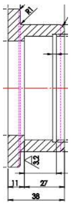
MACHINE TYPE: PK-106 SERIAL NUMBER :M.02246

Woden Pallets Automatic Dispenser



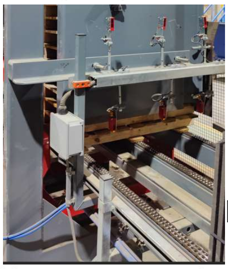


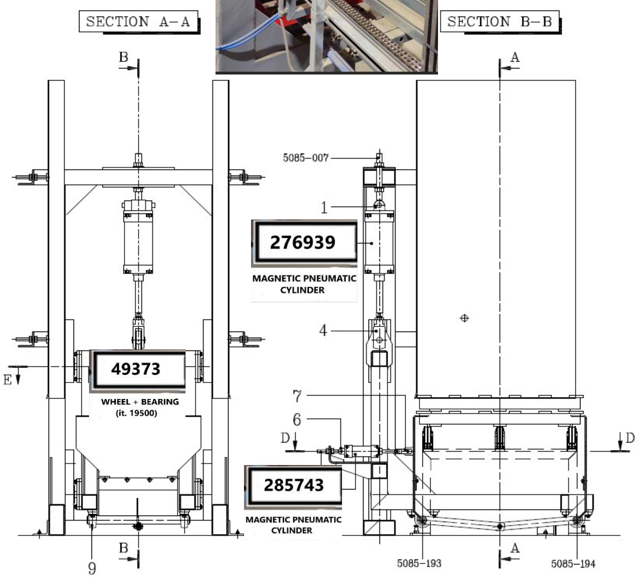




49373

WHEEL + BEARING (it. 19500)





General Description and Function

***This mechanized assembly is responsible for feeding and positioning wooden pallets into the automated line. Although it does not require a formally preventive maintenance program, it is essential to keep its mechanical and pneumatic components in optimal condition to ensure operational continuity and prevent unplanned stoppages.**

Critical Points and Risk Management

Pneumatic System Integrity:
***It is crucial to control air leaks in the 2 main their connections, hoses, and the pressure regulator as these directly affect the efficiency and repeatability of the tilting motion.**

Mechanical Component Wear:
***Monitor wear on the displacement guides of the 4 wheels and the appearance of play in their axles, as well as in the cylinder forks and supports.**

SUPERVISION LEVEL

MONTHLY

49373 Wheel + Bearing (it. 19500) (4 units)
276939 CIL-P SENSDMSG-P050 (1 u)
276940 Fork for pneumatic cylinder (1 u)
276944 Electronic sensor cylinder (1 u)
277217 Scaning photocell (6 units)
285743 CIL-P-MGN-AIRTAC-SAI63X75SG (1 u)
276936 Fork for pneumatic cylinder (1 u)
277176 Solenoid valve (2 units)

Visual/auditory inspection of operation and air leaks.
Supervision of the general state of key components.

Mechanical Verification:

***Check for wear or erosion on the displacement**

***Verify the existence of play in the axles of the 4**

***Inspect for play in the forks and support axles of the 2 pneumatic cylinders.**

Pneumatic Circuit Inspection:

Cylinders:
***Condition of the rod, seals, and mountings.**

Hoses and Connectors:
***Cracks, twists, or leaks.**

Solenoid Valves:
***Correct operation and connection status.**

Pressure Regulator:
***Functional gauge and absence of leaks.**

Key Component Supervision:
***Visual review of the general state of the components listed in the monthly table (sensors, solenoid valves, etc.) to detect physical damage or excessive dirt.**

CLEANING

WEEKLY

***The rolling surfaces of the 4 wheels and their guides to remove dirt, sawdust, or debris that could affect movement.**

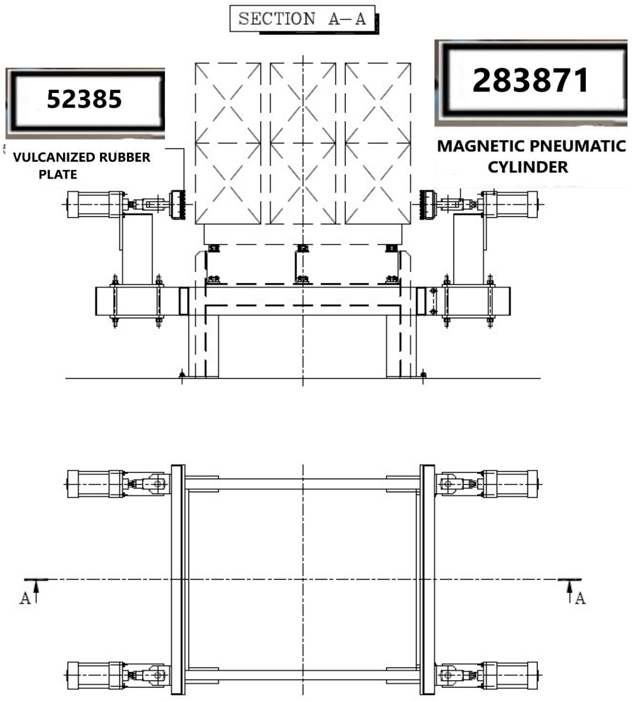
***The horizontal and lateral surfaces of the structure and the cylinders.**

***Use appropriate tools (brushes, dry cloths) to remove dust debris, and any accumulated material.**

AUTOMATIC PALLETIZING LINE

MACHINE TYPE: PK-106 SERIAL NUMBER :M.02246

Squeezer



General Description and Function

***This assembly of 4 pneumatic cylinders, known as the "Squeezer," functions to group and compact the 6 or 3 tiles packages onto the wooden pallet at the final stage of palletizing.**
***Its action involves pushing the 2 or 4 packages previously positioned by the robot in the forming toward the center of forming toward the center of the pallet. This ensures a**
***This ensure stable and compact load for subsequent handling and transport.**

Critical Points and Risk Management

Correct Package Alignment:
***Incorrect arrangement or insufficient compaction of the side packages can cause load shift, resulting in broken tiles during during forklift pickup, on the exit conveyor, or during truck loading. This is the primary operational risk.**

Pneumatic System Integrity:
***Air leaks compromise compaction force and speed, potentially leaving packages loose.**

Condition of Contact Components:
***Wear on the vulcanized rubber plates and play in the mechanisms reduce compaction effectiveness and can damage the packages.**

SUPERVISION LEVEL

DAILY

Operational Visual and Auditory Inspection:

***Listen and look for air leaks (hissing) throughout the entire "Squeezer" assembly.**

***Observe the actuation and full displacement of all 4 cylinders during the compaction cycle, verifying it is uniform and forceful.**

WEEKLY

Visual Verification of Components:

***General condition of the 4 cylinders (mountings, rods).**

***Condition of the compactor support forks.**

***Condition of air connectors and hoses (cracks, twists).**

***Condition of the air pressure regulator (gauge, leaks).**

MONTHLY

52385 Vulcanized rubber plate (4 units)
276856 Fork for pneumatic cylinder (4 units)
276852 Magnetic sensor cylinder (2 units)
277177 Solenoid valve (1 u)
283871 CIL-P-MGN (4 units)

Mechanical and Wear Verification:

Compactor rubber wear:
***Inspect all 4 vulcanized rubber plates for cuts, excessive abrasion, or loss of elasticity.**

Play in forks and support axles:
***Check for excessive play in the 4 forks and their support axles.**

General Component Inspection:
***Review the condition of the 4 cylinders , the solenoid valve, and the associated pneumatic circuit.**

CLEANING

WEEKLY

***Adhesions (stray stretch film, dust, mortar residue) from the "Squeezer" rubber pads (vulcanized plates) to ensure clean uniform contact with the packages.**

***Cylinder surfaces and sides.**

***Use appropriate tools (brushes, dry cloths) to remove dust, debris, and any accumulated material.**